Na poprzednich zajęciach:

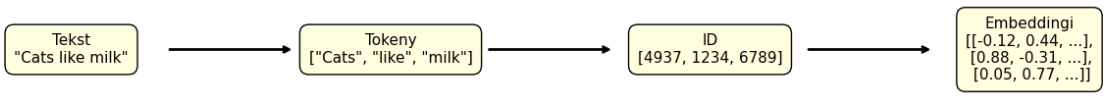
- **tokenizacja** – podział tekstu na słowa / tokeny,
- **spaCy Matcher** – znajdowanie wzorców na podstawie części mowy i cech morfologicznych.

Wektory słów

Tokenizacja a wektory słów

Tokenizacja -> Podziel tekst na kawałki (ID ze słownika)

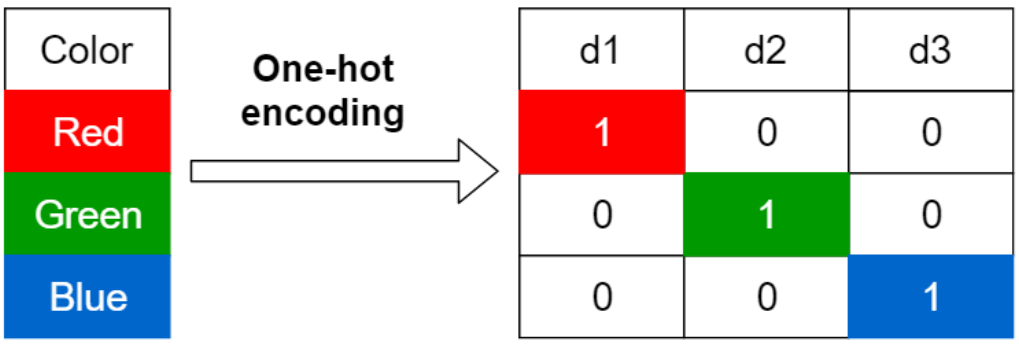
Embedding -> Przedstaw każde słowo w postaci liczb, aby algorytm mógł z nim dalej pracować



Komputer nie „rozumie” tekstu, tylko operuje na liczbach. Zatem każde słowo (np. „król”, „kot”, „miłość”) trzeba jakoś zamienić na liczby, żeby można było nim manipulować w modelu.

Najprostszy pomysł — **one-hot encoding**

```
słownik = [kot, król, królowa]
"król" → [0, 1, 0]
```



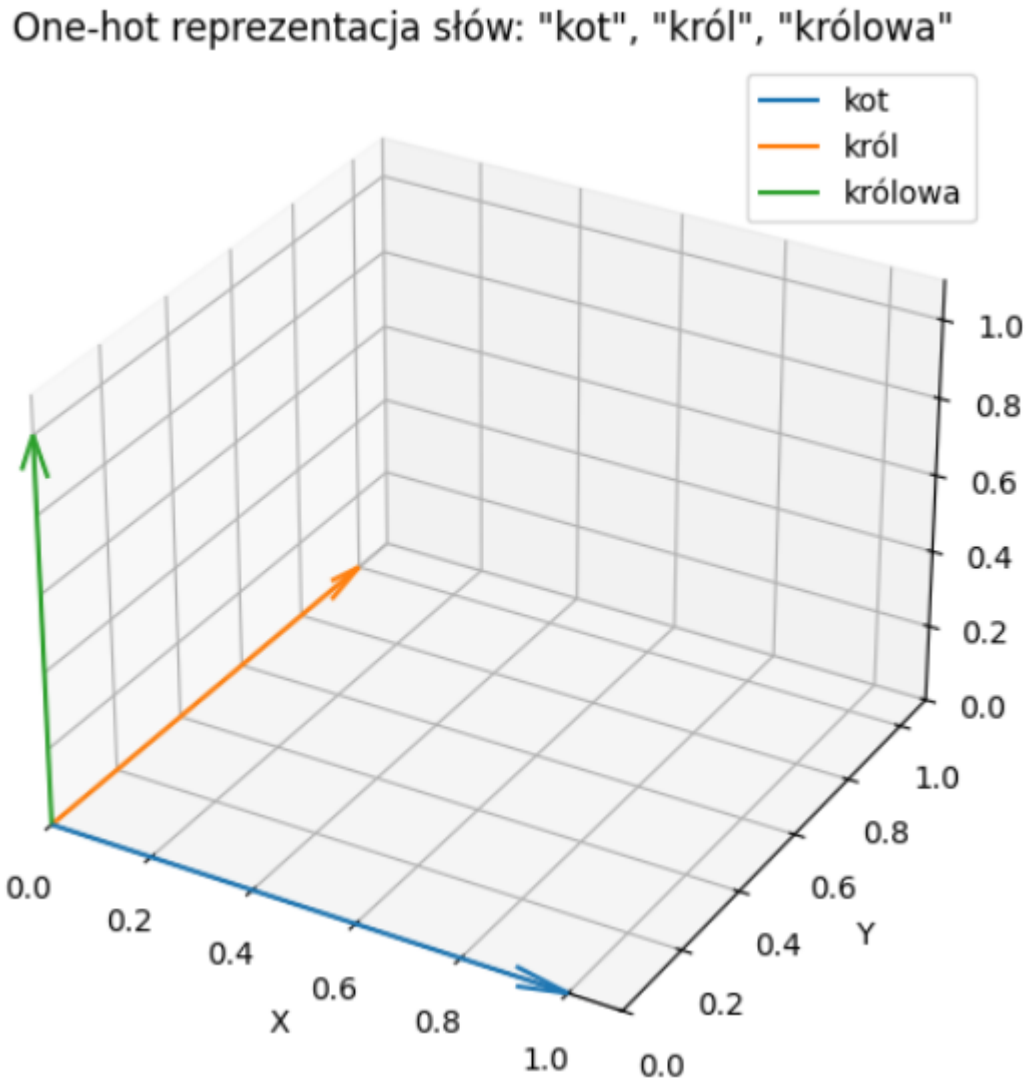
Dlaczego one-hot encoding jest problematyczny?

❌ Bardzo duży wymiar - Jeśli mamy słownik np. 100 000 słów, każdy wektor ma długość 100 000 — to ogromne marnotrawstwo pamięci.

❌ Brak informacji o znaczeniu - Wszystkie wektory są od siebie tak samo odległe — np. “kot” i “pies” są tak samo różne jak “kot” i “samolot”. Nie ma żadnego pojęcia o podobieństwie semantycznym (znaczeniowym).

❌ Brak możliwości uogólniania - Model nie może „zrozumieć”, że pewne słowa mają podobne znaczenie — każde słowo jest całkowicie niezależne.

```
słownik = [kot, król, królowa]
"kot" → [1, 0, 0]
"król" → [0, 1, 0]
"królowa" → [0, 0, 1]
```



Widzimy, że każdy wektor idzie w swoją stronę, każdy jest tak samo odległy od siebie

(90 stopni). Żadne dwa nie są do siebie bliżej ani dalej — odległość między każdym z nich jest taka sama.

W naszym przykładzie: kot, król i królowa są tak samo podobni i różni = nie ma wartości semantycznej.

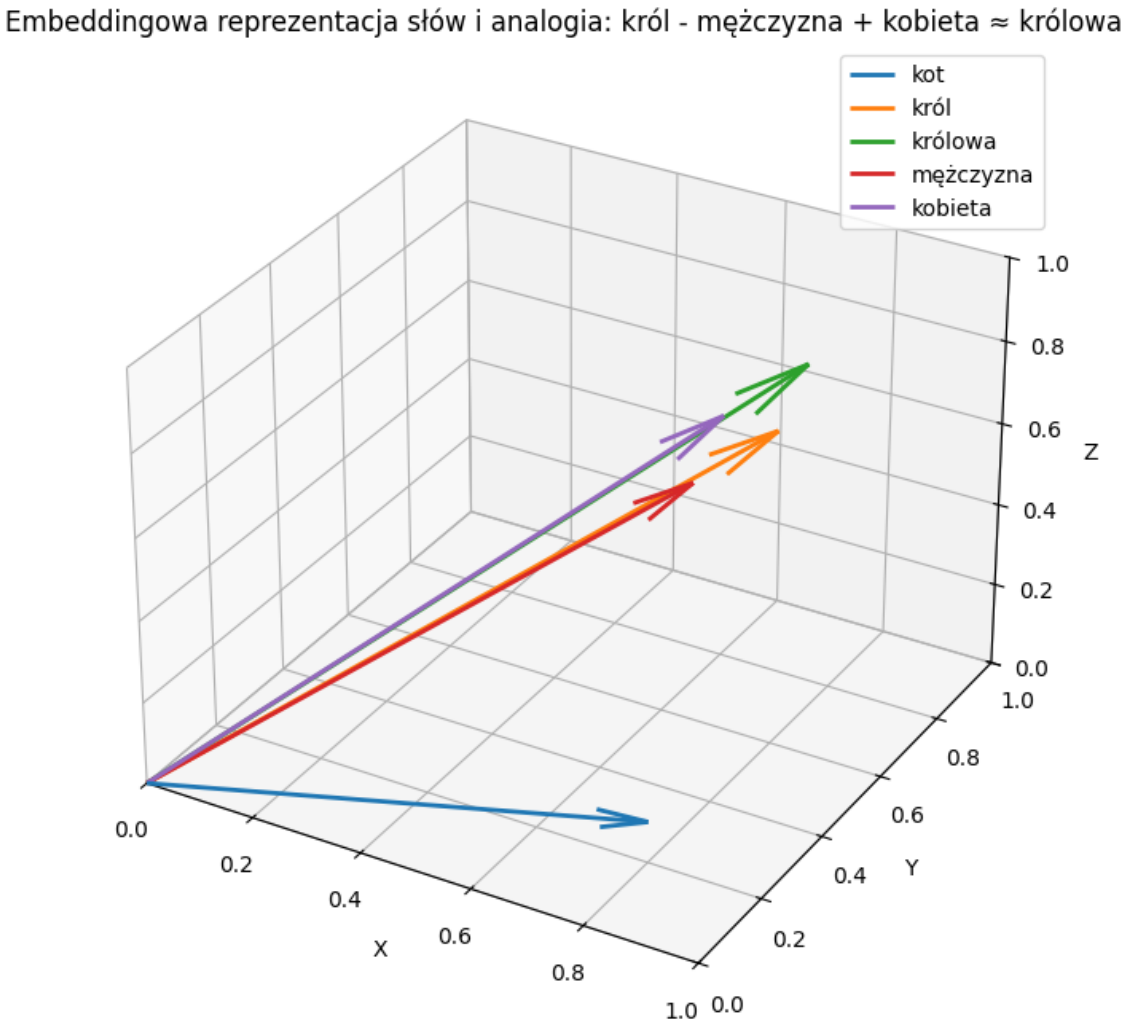
Embedingi

Embeddingi rozwiązują ten problem. W embeddingach uczymy model, żeby słowa, które występują w podobnych kontekstach, miały **podobne wektory**.

Dwa wektory są podobne, kiedy wskazują w podobnym kierunku w przestrzeni wielowymiarowej. Czyli: jeśli narysujesz te wektory jako strzałki zaczynające się w tym samym punkcie (np. w punkcie (0,0,0)) to mają **mały kąt między sobą**.

Przykład:

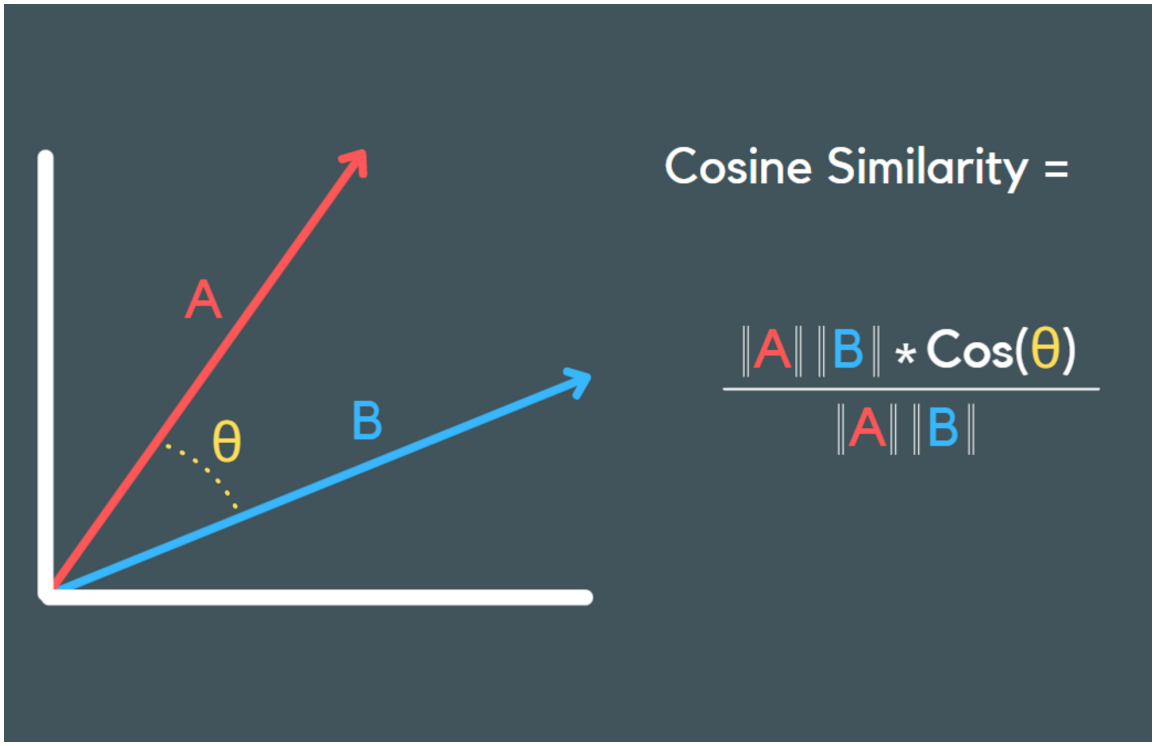
```
"king" → [0.25, 0.88, -0.12, 0.40, ...]
"queen" → [0.22, 0.90, -0.10, 0.45, ...]
"cat" → [-0.70, 0.11, 0.45, 0.20, ...]
```




```
król - mężczyzna + kobieta ≈ królowa
```

Teraz punkty w przestrzeni układają się tak, że słowa o podobnym znaczeniu są blisko siebie. To jest właśnie **semantyczne podobieństwo** — podobieństwo w sensie znaczenia, a nie liter czy indeksów w tabeli.

Cosinusowe podobieństwo — jak mierzymy „bliskość znaczeń”



Cosinusowe podobieństwo to sposób na sprawdzenie, czy dwa wektory są „w tym samym kierunku”:

- 1 → bardzo podobne (np. „kot” i „pies”)
- 0 → brak podobieństwa (np. „kot” i „okno”)
- -1 → całkowite przeciwieństwo

Ważne: Patrzymy nie na długość wektorów, tylko na kąt między nimi. Dwa wektory, które „patrzą” w tym samym kierunku, oznaczają słowa o podobnym znaczeniu.

Skąd się biorą te wektory?

Z treningu modelu językowego (np. Word2Vec, GloVe, FastText, BERT). Model uczy się na dużych korpusach tekstu, obserwując konteksty słów.

Efekt:

„king” i „queen” często występują w podobnych zdaniach (np. royal palace, throne, reign) → mają podobne wektory.

„dog” i „cat” też → inna grupa semantyczna.

Trochę praktyki

Zaczniemy od zapoznania się z biblioteką **gensim** Jej nazwa pochodzi od "GENerate SIMilar" — bo jej głównym celem jest znajdowanie podobieństwa między dokumentami lub słowami.

Użyjemy klasy **KeyedVectors** - to strukturą danych w gensim, która przechowuje wektory słów (embeddingi) wytrenowane dowolnym modelem. Czyli po prostu taki kontener do pracy z wektorami - niezależnie od tego, jak one zostały wytrenowane.

In [1]:
```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x
86_64.whl.metadata (8.4 kB)
Collecting numpy>=1.18.5 (from gensim)
  Downloading numpy-2.5.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x8
6_64.whl.metadata (6.6 kB)
Collecting scipy>=1.7.0 (from gensim)
  Downloading scipy-1.18.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x
86_64.whl.metadata (62 kB)
Collecting smart_open>=1.8.1 (from gensim)
  Downloading smart_open-8.0.3-py3-none-any.whl.metadata (25 kB)
Collecting wrapt (from smart_open>=1.8.1->gensim)
  Downloading wrapt-2.5.1-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_6
4.manylinux_2_5_x86_64.whl.metadata (7.6 kB)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86
_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 62.3 MB/s  0:00:00m
0:00:010:01
Downloading numpy-2.5.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_
64.whl (16.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 16.7/16.7 MB 99.4 MB/s  0:00:00
Downloading scipy-1.18.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86
_64.whl (35.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 35.3/35.3 MB 99.8 MB/s  0:00:00m
eta 0:00:01
Downloading smart_open-8.0.3-py3-none-any.whl (73 kB)
Downloading wrapt-2.5.1-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.m
anylinux_2_5_x86_64.whl (250 kB)
Installing collected packages: wrapt, numpy, smart_open, scipy, gensim
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5/5 [gensim]2m4/5 [gensim]
Successfully installed gensim-4.4.0 numpy-2.5.4 scipy-1.18.1 smart_open-8.0.3
wrapt-2.5.1

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
```

In [2]:
```
import gensim
from gensim.models import KeyedVectors
```

weźmiemy sobie polskie pre-trenowane wektory: http://dsmodels.nlp.ipipan.waw.pl
pobieramy plik:

In [3]:
```
!wget http://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricted-
```

```
7[Files: 0  Bytes: 0  [0 B/s] Re]87[http://dsmodels.nlp.ipipan.waw]87nkjp+wik
i-forms-rest 100% [============================>]     389     --.-KB/s87HTTP
response 301 Moved Permanently [http://dsmodels.nlp.ipipan.waw.pl/dsmodels/nk
jp+wiki-forms-restricted-300-skipg-ns.txt.gz]
87nkjp+wiki-forms-rest 100% [============================>]     389     --.-
KB/s87Enqueue https://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-res
tricted-300-skipg-ns.txt.gz
87[Files: 0  Bytes: 389  [1.06KB/]87WARNING: The certificate is NOT trusted.
The certificate issuer is unknown.
87[https://dsmodels.nlp.ipipan.wa]87Saving 'nkjp+wiki-forms-restricted-300-sk
ipg-ns.txt.gz'
87nkjp+wiki-forms-rest   0% [>                            ]  368.00K    --.-
KB/s87[Files: 0  Bytes: 389  [389 B/s]87nkjp+wiki-forms-rest   0% [>
]    2.46M    2.10MB/s87[Files: 0  Bytes: 389  [194 B/s]87[Files: 0  Bytes: 3
89  [129 B/s]87nkjp+wiki-forms-rest   2% [>                            ]   4
3.64M   21.62MB/s87[Files: 0  Bytes: 389  [97 B/s]]87nkjp+wiki-forms-rest
5% [>                            ]  117.68M   39.09MB/s87[Files: 0  Bytes: 3
89  [77 B/s]]87nkjp+wiki-forms-rest   6% [>                            ]  12
3.44M   30.76MB/s87[Files: 0  Bytes: 389  [64 B/s]]87nkjp+wiki-forms-rest
6% [>                            ]  123.48M   24.61MB/s87[Files: 0  Bytes: 3
89  [55 B/s]]87nkjp+wiki-forms-rest   8% [=>                           ]  16
7.69M   27.88MB/s87[Files: 0  Bytes: 389  [48 B/s]]87nkjp+wiki-forms-rest
8% [=>                           ]  173.27M   24.69MB/s87[Files: 0  Bytes: 3
89  [43 B/s]]87nkjp+wiki-forms-rest  11% [==>                          ]  22
7.88M   28.43MB/s87[Files: 0  Bytes: 389  [38 B/s]]87nkjp+wiki-forms-rest  1
4% [===>                         ]  299.61M   33.24MB/s87[Files: 0  Bytes: 3
89  [35 B/s]]87nkjp+wiki-forms-rest  18% [====>                        ]  38
1.93M   38.14MB/s87[Files: 0  Bytes: 389  [32 B/s]]87nkjp+wiki-forms-rest  2
2% [=====>                       ]  461.01M   41.87MB/s87[Files: 0  Bytes: 3
89  [29 B/s]]87nkjp+wiki-forms-rest  22% [=====>                       ]  47
1.82M   39.28MB/s87[Files: 0  Bytes: 389  [27 B/s]]87nkjp+wiki-forms-rest  2
2% [=====>                       ]  472.16M   36.28MB/s87[Files: 0  Bytes: 3
89  [25 B/s]]87nkjp+wiki-forms-rest  26% [======>                      ]  54
0.82M   38.59MB/s87[Files: 0  Bytes: 389  [24 B/s]]87nkjp+wiki-forms-rest  2
9% [=======>                     ]  596.34M   39.72MB/s87[Files: 0  Bytes: 3
89  [22 B/s]]87nkjp+wiki-forms-rest  32% [========>                    ]  66
3.66M   41.44MB/s87[Files: 0  Bytes: 389  [21 B/s]]87nkjp+wiki-forms-rest  3
5% [=========>                   ]  732.23M   43.04MB/s87[Files: 0  Bytes: 3
89  [20 B/s]]87nkjp+wiki-forms-rest  39% [==========>                  ]  80
4.65M   44.67MB/s87[Files: 0  Bytes: 389  [19 B/s]]87nkjp+wiki-forms-rest  4
2% [===========>                 ]  865.97M   45.55MB/s87[Files: 0  Bytes: 3
89  [18 B/s]]87nkjp+wiki-forms-rest  44% [============>                ]  91
8.28M   45.88MB/s87[Files: 0  Bytes: 389  [17 B/s]]87nkjp+wiki-forms-rest  4
7% [=============>               ]  983.46M   46.80MB/s87[Files: 0  Bytes: 3
89  [16 B/s]]87nkjp+wiki-forms-rest  49% [=============>               ]
1017M   46.19MB/s87[Files: 0  Bytes: 389  [16 B/s]]87nkjp+wiki-forms-rest  5
2% [==============>              ]    1.06G   47.21MB/s87[Files: 0  Bytes: 3
89  [15 B/s]]87nkjp+wiki-forms-rest  56% [===============>             ]
1.12G   49.89MB/s87[Files: 0  Bytes: 389  [14 B/s]]87nkjp+wiki-forms-rest  5
8% [================>            ]    1.17G   50.54MB/s87[Files: 0  Bytes: 3
89  [14 B/s]]87nkjp+wiki-forms-rest  61% [=================>           ]
1.23G   49.90MB/s87[Files: 0  Bytes: 389  [13 B/s]]87nkjp+wiki-forms-rest  6
4% [==================>          ]    1.29G   52.49MB/s87[Files: 0  Bytes: 3
89  [13 B/s]]87nkjp+wiki-forms-rest  68% [===================>         ]
1.36G   55.47MB/s87[Files: 0  Bytes: 389  [12 B/s]]87nkjp+wiki-forms-rest  7
```

```
1% [====================>         ]    1.44G   56.88MB/s87[Files: 0  Bytes: 3
89  [12 B/s]]87nkjp+wiki-forms-rest  75% [=====================>        ]
1.50G   59.58MB/s87[Files: 0  Bytes: 389  [12 B/s]]87nkjp+wiki-forms-rest  7
8% [======================>       ]    1.58G   60.57MB/s87[Files: 0  Bytes: 3
89  [11 B/s]]87nkjp+wiki-forms-rest  82% [=======================>      ]
1.65G   60.76MB/s87[Files: 0  Bytes: 389  [11 B/s]]87nkjp+wiki-forms-rest  8
6% [========================>     ]    1.73G   60.61MB/s87[Files: 0  Bytes: 3
89  [11 B/s]]87nkjp+wiki-forms-rest  89% [========================>     ]
1.79G   59.91MB/s87[Files: 0  Bytes: 389  [10 B/s]]87nkjp+wiki-forms-rest  9
1% [=========================>    ]    1.83G   61.42MB/s87[Files: 0  Bytes: 3
89  [10 B/s]]87nkjp+wiki-forms-rest  95% [==========================>   ]
1.91G   64.80MB/s87[Files: 0  Bytes: 389  [10 B/s]]87nkjp+wiki-forms-rest  9
6% [===========================> ]    1.93G   62.82MB/s87[Files: 0  Bytes: 3
89  [9 B/s] ]87nkjp+wiki-forms-rest  96% [===========================> ]
1.93G   60.41MB/s87[Files: 0  Bytes: 389  [9 B/s] ]87nkjp+wiki-forms-rest  9
8% [===========================> ]    1.98G   59.48MB/s87nkjp+wiki-forms-res
t 100% [============================>]    2.00G   58.72MB/s87HTTP response 2
00 OK [https://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricte
d-300-skipg-ns.txt.gz]
87nkjp+wiki-forms-rest 100% [============================>]    2.00G   58.72
MB/s87[Files: 1  Bytes: 2.00G [50.69M]8
```

Rozpakowujemy dane

In [4]:
```
!gzip -d "nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz"
```

Badamy format pliku

In [102…
```
#!head -n 100 nkjp+wiki-forms-restricted-300-skipg-ns.txt
#Pierwsza linia - lość słów (1982479), rozmiar wektora(300)
#Każda kolejna linia - słowo, liczby reperezentujące wektor
```

(to może chwilę zająć)

In [6]:
```
#Ładowanie wektorów z pliku do naszego KeyedVectors
wv = KeyedVectors.load_word2vec_format('nkjp+wiki-forms-restricted-300-skipg
```

In [7]:
```
wv.key_to_index.keys()
wv['dom']
```

Out[7]:
```
array([-0.39294 , -0.156333,  0.017816, -0.208752,  0.442839, -0.389071,
       -0.22354 ,  0.119761,  0.201861,  0.136966,  0.005829, -0.036992,
       -0.075693, -0.424958,  0.031507, -0.214998, -0.11862 , -0.059799,
       -0.150971, -0.176567,  0.444136,  0.18925 ,  0.241265, -0.248733,
       -0.176206,  0.579075,  0.031083,  0.253453,  0.101754,  0.144021,
       -0.169662, -0.005122,  0.056225,  0.078038, -0.557848, -0.365922,
       -0.108863,  0.322074, -0.178232,  0.289982, -0.205699,  0.001991,
        0.001029, -0.074944,  0.173145,  0.044346, -0.032673, -0.060124,
        0.152098, -0.293121, -0.280878,  0.377744,  0.402627, -0.737609,
        0.020663, -0.01301 ,  0.134813, -0.052369,  0.046645, -0.024955,
        0.100392, -0.295437, -0.418354,  0.396864,  0.232666,  0.022819,
       -0.018474,  0.385309, -0.084371,  0.269105,  0.149824, -0.161652,
        0.282115,  0.177849, -0.241524, -0.240634, -0.111456,  0.070668,
       -0.216499, -0.139156, -0.576622, -0.136423,  0.403575,  0.155968,
        0.218948, -0.002413,  0.123077,  0.102393,  0.741444,  0.390994,
       -0.110964,  0.134329,  0.265171,  0.251443, -0.221384, -0.344603,
        0.283788,  0.284915,  0.057016,  0.384028, -0.733822,  0.303532,
       -0.583847, -0.02494 , -0.218198,  0.018208,  0.145667,  0.255729,
        0.498478, -0.112536, -0.047852, -0.337887,  0.031235, -0.432209,
        0.208977,  0.012221,  0.19098 , -0.206233, -0.113016,  0.111511,
       -0.059505,  0.027593, -0.100111, -0.230436, -0.078137,  0.278976,
        0.19794 ,  0.210459, -0.155359, -0.112397,  0.26439 , -0.199999,
        0.058368,  0.032107,  0.110054,  0.051876, -0.240713,  0.277577,
        0.289985, -0.200094, -0.101406,  0.131032, -0.059501,  0.134268,
        0.148928, -0.085373,  0.030669,  0.068627, -0.215063, -0.081981,
       -0.052877, -0.291941, -0.458875,  0.050618,  0.214273,  0.184849,
       -0.038208,  0.036236, -0.642125, -0.464941,  0.372085, -0.187011,
       -0.069729,  0.035051,  0.099067,  0.057524,  0.006579,  0.533439,
       -0.198236, -0.344194, -0.067796, -0.063289, -0.069952, -0.041706,
        0.142563,  0.05342 , -0.054871,  0.2577  ,  0.50673 ,  0.208496,
        0.650207, -0.252643,  0.320542, -0.54734 , -0.637634, -0.51038 ,
        0.025401,  0.05803 ,  0.16656 ,  0.257388,  0.305958, -0.219199,
        0.2086  , -0.251887, -0.185647, -0.027838, -0.186104, -0.070295,
       -0.094713,  0.102284, -0.123802,  0.083343,  0.315851, -0.031123,
        0.238093,  0.149717, -0.004316, -0.181932, -0.086956,  0.531083,
        0.112422,  0.231151,  0.242227,  0.012977, -0.095725, -0.283862,
       -0.061226,  0.311026,  0.146489,  0.054716, -0.048658, -0.282848,
       -0.054836, -0.030167, -0.090273, -0.435833, -0.287074, -0.351567,
       -0.002671, -0.020709, -0.276286,  0.281762, -0.114932, -0.243566,
       -0.188988,  0.007132,  0.146958, -0.269132, -0.034678,  0.164219,
       -0.090405,  0.044055, -0.405617,  0.162068,  0.008641,  0.142725,
       -0.065804,  0.136329,  0.247081, -0.144136, -0.007592,  0.113807,
        0.14899 , -0.012305, -0.346247,  0.111213,  0.011855,  0.250259,
        0.068319,  0.100238,  0.010446,  0.127471, -0.033004, -0.079141,
       -0.531317,  0.318433,  0.562701,  0.354442,  0.001789,  0.171425,
        0.010197,  0.043681, -0.031715,  0.505568, -0.453908,  0.050882,
       -0.294158, -0.279005, -0.121981,  0.182517,  0.419662,  0.325643,
       -0.027132, -0.020383,  0.133923, -0.047128,  0.406531, -0.385169,
       -0.148565, -0.109004, -0.062878, -0.236207,  0.008543,  0.044215,
        0.216761,  0.017197, -0.034889,  0.311272,  0.130243, -0.122846],
      dtype=float32)
```

Znajdziemy najbardziej podobne słowo do słowa kobieta.

In [66]:
```
wv.most_similar('kobieta')
```

Out[66]:
```
[('mężczyzna', 0.7043285369873047),
 ('dziewczyna', 0.6813511252403259),
 ('ciężarna', 0.612939715385437),
 ('dziewczynka', 0.6094770431518555),
 ('50-letnia', 0.5805984139442444),
 ('młoda', 0.5788260102272034),
 ('40-letnia', 0.5631040930747986),
 ('kobietą', 0.5611501336097717),
 ('dama', 0.5526077747344971),
 ('mężczyzną', 0.5519220232963562)]
```

Znajdź najbardziej podobne słowo do słowa "kobieta", które nie jest podobne do słowa "mężczyzna"

In [61]:
```
wv.most_similar(positive=['kobieta'], negative=['mężczyzna'])
```

Out[61]:
```
[('Teresa', 0.3597542345523834),
 ('Irlandka', 0.35389503836631775),
 ('Barbara', 0.33941537141799927),
 ('Helena', 0.33837634325027466),
 ('Eleonora', 0.32829549908638),
 ('mianowana', 0.3252837657928467),
 ('Maryse', 0.32269802689552307),
 ('Emmanuella', 0.32241636514663696),
 ('czarnoskóra', 0.3224010169506073),
 ('Izabela', 0.3222866654396057)]
```

W grupie "pies kot żubr budyń" znajdź słowo, które nie pasuje.

In [26]:
```
wv.doesnt_match(['pies', 'kot', 'żubr', 'budyń'])
```

Out[26]:
```
'budyń'
```

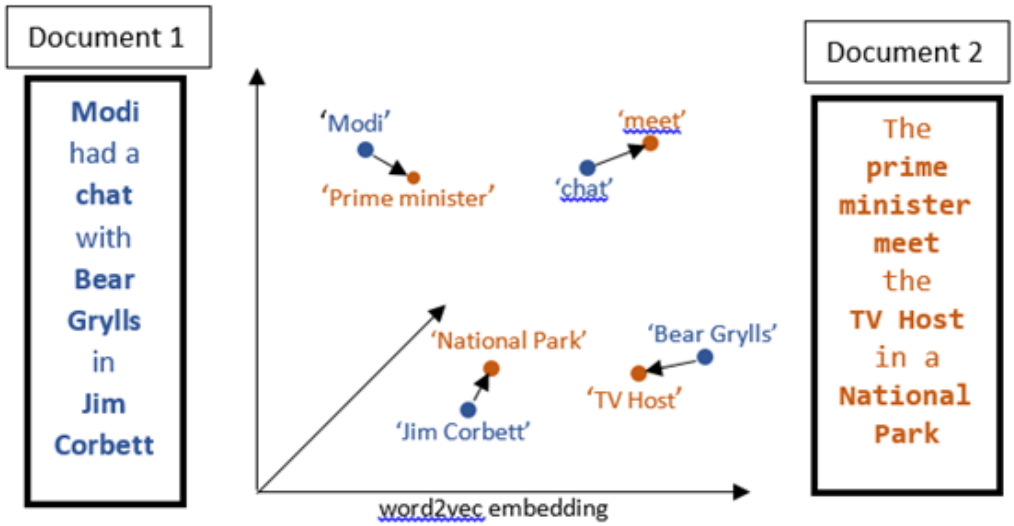
zbadaj podobieństwo między kotem a psem

In [11]:
```
wv.similarity('kot', 'pies')
```

Out[11]:
```
np.float32(0.6970217)
```

Teraz zajmiemy się podobieństwami pomiędzy zdaniami

Word mover distance to minimalna suma odległości, jaką trzeba „pokonać”, aby słowa z jednego zdania pokryły znaczeniowo słowa z drugiego zdania.



In [12]:
```
!pip install POT #Po tej instalacji gensim może szybciej i dokładniej liczyć
```

```
Collecting POT
  Downloading pot-0.9.7.post1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_2
8_x86_64.whl.metadata (44 kB)
Requirement already satisfied: numpy>=1.16 in ./venv/lib64/python3.12/site-pa
ckages (from POT) (2.5.4)
Requirement already satisfied: scipy>=1.6 in ./venv/lib64/python3.12/site-pac
kages (from POT) (1.18.1)
Downloading pot-0.9.7.post1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_
x86_64.whl (32.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 32.9/32.9 MB 78.6 MB/s  0:00:00 e
ta 0:00:01
Installing collected packages: POT
Successfully installed POT-0.9.7.post1

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
```

In [13]:
```
sentence_1 = 'Byłem dzisiaj w kinie'.lower().split()
sentence_2 = 'Poszłam wczoraj do teatru'.lower().split()
```

In [14]:
```
wv.wmdistance(sentence_1, sentence_2)
```

Out[14]:
```
1.176677407878612
```

WMD nie jest ograniczone do 0-1, ale przy normalizacji embeddingów zwykle wartości mieszczą się w 0-4 dla krótkich zdań.

wynik ~1.25 sugeruje, że zdania są umiarkowanie podobne semantycznie.

Dla porównania:

- Dwa identyczne zdania → WMD ≈ 0
- Zupełnie różne zdania → WMD > 2.5-3

Flair

Flair to biblioteka NLP od Zalando Research (Python)

Główna idea:

- Klasyczne embeddingi (Word2Vec, GloVe, FastText) dają stały wektor dla każdego słowa, niezależnie od kontekstu:

  "bank" w *"river bank"* i *"money bank"* → ten sam wektor.

- Flair tworzy kontekstowe embeddingi(wektor słowa zależy od jego zdania / kontekstu)

  więc "*bank*" w "*river bank*" będzie inny niż w "*money bank*".

Flair uczy się tego przez modele LSTM działające w dwóch kierunkach (forward i backward).

In [15]:
```
!pip install flair
```

```
Collecting flair
  Downloading flair-0.15.1-py3-none-any.whl.metadata (12 kB)
Collecting boto3>=1.20.27 (from flair)
  Downloading boto3-1.43.114-py3-none-any.whl.metadata (6.6 kB)
Collecting conllu<5.0.0,>=4.0 (from flair)
  Downloading conllu-4.5.3-py2.py3-none-any.whl.metadata (19 kB)
Collecting deprecated>=1.2.13 (from flair)
  Downloading deprecated-3.0.0-py3-none-any.whl.metadata (6.5 kB)
Collecting ftfy>=6.1.0 (from flair)
  Downloading ftfy-6.3.1-py3-none-any.whl.metadata (7.3 kB)
Collecting gdown>=4.4.0 (from flair)
  Downloading gdown-6.4.2-py3-none-any.whl.metadata (11 kB)
Collecting huggingface-hub>=0.10.0 (from flair)
  Downloading huggingface_hub-2.2.0-py3-none-any.whl.metadata (16 kB)
Collecting langdetect>=1.0.9 (from flair)
  Downloading langdetect-1.0.9.tar.gz (981 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 981.5/981.5 kB 11.3 MB/s  0:0
0:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting lxml>=4.8.0 (from flair)
  Downloading lxml-6.1.3-cp312-cp312-manylinux_2_26_x86_64.manylinux_2_28_x86
_64.whl.metadata (3.3 kB)
Collecting matplotlib>=2.2.3 (from flair)
  Downloading matplotlib-3.11.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_
17_x86_64.whl.metadata (80 kB)
Collecting more-itertools>=8.13.0 (from flair)
  Downloading more_itertools-11.2.1-py3-none-any.whl.metadata (42 kB)
Collecting mpld3>=0.3 (from flair)
  Downloading mpld3-0.5.12-py3-none-any.whl.metadata (5.3 kB)
Collecting pptree>=3.1 (from flair)
  Downloading pptree-3.1.tar.gz (3.0 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: python-dateutil>=2.8.2 in ./venv/lib64/python
3.12/site-packages (from flair) (2.9.0.post0)
Collecting pytorch-revgrad>=0.2.0 (from flair)
  Downloading pytorch_revgrad-0.2.0-py3-none-any.whl.metadata (1.7 kB)
Collecting regex>=2022.1.18 (from flair)
  Using cached regex-2026.9.29-cp312-cp312-manylinux2014_x86_64.manylinux_2_1
7_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)
Collecting scikit-learn>=1.0.2 (from flair)
  Downloading scikit_learn-1.9.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_
2_28_x86_64.whl.metadata (9.3 kB)
Collecting segtok>=1.5.11 (from flair)
  Downloading segtok-1.5.11-py3-none-any.whl.metadata (9.0 kB)
Collecting sqlitedict>=2.0.0 (from flair)
  Downloading sqlitedict-2.1.0.tar.gz (21 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting tabulate>=0.8.10 (from flair)
  Downloading tabulate-0.10.0-py3-none-any.whl.metadata (40 kB)
Requirement already satisfied: torch>=1.13.1 in ./venv/lib64/python3.12/site-
```

```
packages (from flair) (2.14.1)
Collecting tqdm>=4.63.0 (from flair)
  Using cached tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Collecting transformer-smaller-training-vocab>=0.2.3 (from flair)
  Downloading transformer_smaller_training_vocab-0.4.2-py3-none-any.whl.metad
ata (7.9 kB)
Collecting transformers<5.0.0,>=4.25.0 (from transformers[sentencepiece]<
5.0.0,>=4.25.0->flair)
  Downloading transformers-4.57.6-py3-none-any.whl.metadata (43 kB)
Collecting wikipedia-api>=0.5.7 (from flair)
  Downloading wikipedia_api-0.16.0-py3-none-any.whl.metadata (44 kB)
Collecting bioc<3.0.0,>=2.0.0 (from flair)
  Downloading bioc-2.1-py3-none-any.whl.metadata (4.6 kB)
Collecting jsonlines>=1.2.0 (from bioc<3.0.0,>=2.0.0->flair)
  Downloading jsonlines-4.0.0-py3-none-any.whl.metadata (1.6 kB)
Collecting intervaltree (from bioc<3.0.0,>=2.0.0->flair)
  Downloading intervaltree-3.2.1-py2.py3-none-any.whl.metadata (12 kB)
Collecting docopt (from bioc<3.0.0,>=2.0.0->flair)
  Downloading docopt-0.6.2.tar.gz (25 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: filelock in ./venv/lib64/python3.12/site-packa
ges (from transformers<5.0.0,>=4.25.0->transformers[sentencepiece]<5.0.0,>=
4.25.0->flair) (4.2.0)
Collecting huggingface-hub>=0.10.0 (from flair)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: numpy>=1.17 in ./venv/lib64/python3.12/site-pa
ckages (from transformers<5.0.0,>=4.25.0->transformers[sentencepiece]<5.0.0,>
=4.25.0->flair) (2.5.4)
Requirement already satisfied: packaging>=20.0 in ./venv/lib64/python3.12/sit
e-packages (from transformers<5.0.0,>=4.25.0->transformers[sentencepiece]<
5.0.0,>=4.25.0->flair) (26.3)
Collecting pyyaml>=5.1 (from transformers<5.0.0,>=4.25.0->transformers[senten
cepiece]<5.0.0,>=4.25.0->flair)
  Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting requests (from transformers<5.0.0,>=4.25.0->transformers[sentencep
iece]<5.0.0,>=4.25.0->flair)
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers<5.0.0,>=4.25.0->tra
nsformers[sentencepiece]<5.0.0,>=4.25.0->flair)
  Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014
_x86_64.whl.metadata (7.3 kB)
Collecting safetensors>=0.4.3 (from transformers<5.0.0,>=4.25.0->transformers
[sentencepiece]<5.0.0,>=4.25.0->flair)
  Using cached safetensors-0.8.0-cp310-abi3-manylinux_2_17_x86_64.manylinux20
14_x86_64.whl.metadata (4.2 kB)
Requirement already satisfied: fsspec>=2023.5.0 in ./venv/lib64/python3.12/si
te-packages (from huggingface-hub>=0.10.0->flair) (2026.9.0)
Collecting hf-xet<2.0.0,>=1.1.3 (from huggingface-hub>=0.10.0->flair)
  Using cached hf_xet-1.7.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86
_64.whl.metadata (4.9 kB)
Requirement already satisfied: typing-extensions>=3.7.4.3 in ./venv/lib64/pyt
hon3.12/site-packages (from huggingface-hub>=0.10.0->flair) (4.16.0)
Collecting sentencepiece!=0.1.92,>=0.1.91 (from transformers[sentencepiece]<
```

```
5.0.0,>=4.25.0->flair)
  Downloading sentencepiece-0.2.2-cp312-cp312-manylinux_2_27_x86_64.manylinux
_2_28_x86_64.whl.metadata (33 kB)
Collecting protobuf (from transformers[sentencepiece]<5.0.0,>=4.25.0->flair)
  Downloading protobuf-7.36.2-cp310-abi3-manylinux2014_x86_64.whl.metadata (5
95 bytes)
Collecting botocore<1.44.0,>=1.43.114 (from boto3>=1.20.27->flair)
  Downloading botocore-1.43.114-py3-none-any.whl.metadata (5.6 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3>=1.20.27->flair)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.20.0,>=0.19.0 (from boto3>=1.20.27->flair)
  Downloading s3transfer-0.19.2-py3-none-any.whl.metadata (1.7 kB)
Collecting urllib3!=2.2.0,<3,>=1.25.4 (from botocore<1.44.0,>=1.43.114->boto3
>=1.20.27->flair)
  Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Requirement already satisfied: six>=1.5 in ./venv/lib64/python3.12/site-packa
ges (from python-dateutil>=2.8.2->flair) (1.17.0)
Requirement already satisfied: wrapt<3,>=1.16 in ./venv/lib64/python3.12/sit
e-packages (from deprecated>=1.2.13->flair) (2.5.1)
Requirement already satisfied: wcwidth in ./venv/lib64/python3.12/site-packag
es (from ftfy>=6.1.0->flair) (0.9.2)
Collecting beautifulsoup4 (from gdown>=4.4.0->flair)
  Using cached beautifulsoup4-4.15.0-py3-none-any.whl.metadata (3.8 kB)
Collecting attrs>=19.2.0 (from jsonlines>=1.2.0->bioc<3.0.0,>=2.0.0->flair)
  Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting contourpy>=1.0.1 (from matplotlib>=2.2.3->flair)
  Downloading contourpy-1.4.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_2
8_x86_64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib>=2.2.3->flair)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib>=2.2.3->flair)
  Downloading fonttools-4.67.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_1
7_x86_64.whl.metadata (137 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib>=2.2.3->flair)
  Downloading kiwisolver-1.5.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_1
7_x86_64.whl.metadata (5.2 kB)
Collecting pillow>=9 (from matplotlib>=2.2.3->flair)
  Downloading pillow-12.3.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_
x86_64.whl.metadata (9.1 kB)
Collecting pyparsing>=3 (from matplotlib>=2.2.3->flair)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: jinja2 in ./venv/lib64/python3.12/site-package
s (from mpld3>=0.3->flair) (3.1.6)
Requirement already satisfied: scipy>=1.10.0 in ./venv/lib64/python3.12/site-
packages (from scikit-learn>=1.0.2->flair) (1.18.1)
Collecting joblib>=1.4.0 (from scikit-learn>=1.0.2->flair)
  Using cached joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting narwhals>=2.0.1 (from scikit-learn>=1.0.2->flair)
  Downloading narwhals-2.27.1-py3-none-any.whl.metadata (15 kB)
Collecting threadpoolctl>=3.5.0 (from scikit-learn>=1.0.2->flair)
  Downloading threadpoolctl-3.7.0-py3-none-any.whl.metadata (24 kB)
Collecting cloudpickle>=3.0 (from joblib>=1.4.0->scikit-learn>=1.0.2->flair)
  Using cached cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: setuptools>=77.0.3 in ./venv/lib64/python3.12/
site-packages (from torch>=1.13.1->flair) (84.0.0)
Requirement already satisfied: sympy>=1.13.3 in ./venv/lib64/python3.12/site-
```

```
packages (from torch>=1.13.1->flair) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in ./venv/lib64/python3.12/sit
e-packages (from torch>=1.13.1->flair) (3.7)
Requirement already satisfied: cuda-toolkit==13.0.3 in ./venv/lib64/python3.1
2/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cu
solver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->to
rch>=1.13.1->flair) (13.0.3.0)
Requirement already satisfied: cuda-bindings<14,>=13.0.3 in ./venv/lib64/pyth
on3.12/site-packages (from torch>=1.13.1->flair) (13.4.4)
Requirement already satisfied: nvidia-cudnn-cu13==9.24.0.43 in ./venv/lib64/p
ython3.12/site-packages (from torch>=1.13.1->flair) (9.24.0.43)
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.1 in ./venv/lib64/
python3.12/site-packages (from torch>=1.13.1->flair) (0.8.1)
Requirement already satisfied: nvidia-nccl-cu13==2.30.7 in ./venv/lib64/pytho
n3.12/site-packages (from torch>=1.13.1->flair) (2.30.7)
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in ./venv/lib64/pyt
hon3.12/site-packages (from torch>=1.13.1->flair) (3.4.5)
Requirement already satisfied: triton~=3.8.0 in ./venv/lib64/python3.12/site-
packages (from torch>=1.13.1->flair) (3.8.0)
Requirement already satisfied: nvidia-cublas==13.1.1.3.* in ./venv/lib64/pyth
on3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,cura
nd,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linu
x"->torch>=1.13.1->flair) (13.1.1.3)
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in ./venv/lib64/p
ython3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,c
urand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Li
nux"->torch>=1.13.1->flair) (13.0.88)
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in ./venv/lib6
4/python3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupt
i,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system ==
"Linux"->torch>=1.13.1->flair) (13.0.96)
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in ./venv/lib64/pyth
on3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,cura
nd,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linu
x"->torch>=1.13.1->flair) (12.0.0.61)
Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in ./venv/lib64/
python3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupt
i,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system ==
"Linux"->torch>=1.13.1->flair) (13.4.92)
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in ./venv/lib64/pyth
on3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,cura
nd,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linu
x"->torch>=1.13.1->flair) (1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in ./venv/lib64/p
ython3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,c
urand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Li
nux"->torch>=1.13.1->flair) (13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in ./venv/lib64/pyt
hon3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,cur
and,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linu
x"->torch>=1.13.1->flair) (10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in ./venv/lib64/p
ython3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,c
urand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Li
nux"->torch>=1.13.1->flair) (12.0.4.66)
Requirement already satisfied: nvidia-cusparse==12.6.3.3.* in ./venv/lib64/py
```

```
thon3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,cu
rand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Lin
ux"->torch>=1.13.1->flair) (12.6.3.3)
Requirement already satisfied: nvidia-nvtx==13.0.85.* in ./venv/lib64/python
3.12/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curan
d,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linu
x"->torch>=1.13.1->flair) (13.0.85)
Requirement already satisfied: cuda-pathfinder>=1.4.2 in ./venv/lib64/python
3.12/site-packages (from cuda-bindings<14,>=13.0.3->torch>=1.13.1->flair)
(1.8.3)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in ./venv/lib64/python3.12/
site-packages (from sympy>=1.13.3->torch>=1.13.1->flair) (1.3.0)
Collecting accelerate>=0.26.0 (from transformers<5.0.0,>=4.25.0->transformers
[sentencepiece]<5.0.0,>=4.25.0->flair)
  Downloading accelerate-1.15.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: psutil in ./venv/lib64/python3.12/site-package
s (from accelerate>=0.26.0->transformers<5.0.0,>=4.25.0->transformers[sentenc
epiece]<5.0.0,>=4.25.0->flair) (7.2.2)
Collecting click==8.5.0 (from wikipedia-api>=0.5.7->flair)
  Using cached click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting httpx==0.28.1 (from wikipedia-api>=0.5.7->flair)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting tenacity==9.1.4 (from wikipedia-api>=0.5.7->flair)
  Downloading tenacity-9.1.4-py3-none-any.whl.metadata (1.2 kB)
Collecting anyio (from httpx==0.28.1->wikipedia-api>=0.5.7->flair)
  Using cached anyio-4.15.1-py3-none-any.whl.metadata (4.7 kB)
Collecting certifi (from httpx==0.28.1->wikipedia-api>=0.5.7->flair)
  Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx==0.28.1->wikipedia-api>=0.5.7->flair)
  Using cached httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx==0.28.1->wikipedia-api>=0.5.7->flair)
  Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx==0.28.1->wikipedia-api>=
0.5.7->flair)
  Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting soupsieve>=1.6.1 (from beautifulsoup4->gdown>=4.4.0->flair)
  Downloading soupsieve-3.0.3-py3-none-any.whl.metadata (4.4 kB)
Collecting sortedcontainers (from intervaltree->bioc<3.0.0,>=2.0.0->flair)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: MarkupSafe>=2.0 in ./venv/lib64/python3.12/sit
e-packages (from jinja2->mpld3>=0.3->flair) (3.0.4)
Collecting charset_normalizer<4,>=2 (from requests->transformers<5.0.0,>=4.2
5.0->transformers[sentencepiece]<5.0.0,>=4.25.0->flair)
  Using cached charset_normalizer-3.5.2-cp312-cp312-manylinux2014_x86_64.many
linux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (46 kB)
Collecting PySocks!=1.5.7,>=1.5.6 (from requests[socks]->gdown>=4.4.0->flair)
  Downloading PySocks-1.7.1-py3-none-any.whl.metadata (13 kB)
Downloading flair-0.15.1-py3-none-any.whl (1.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 24.8 MB/s  0:00:00
Downloading bioc-2.1-py3-none-any.whl (33 kB)
Downloading conllu-4.5.3-py2.py3-none-any.whl (16 kB)
Downloading transformers-4.57.6-py3-none-any.whl (12.0 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.0/12.0 MB 75.0 MB/s  0:00:00
Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 566.4/566.4 kB 52.6 MB/s  0:00:00
Using cached hf_xet-1.7.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86_6
```

```
4.whl (4.2 MB)
Downloading tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x
86_64.whl (3.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 34.0 MB/s  0:00:00
Downloading boto3-1.43.114-py3-none-any.whl (140 kB)
Downloading botocore-1.43.114-py3-none-any.whl (16.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 16.1/16.1 MB 105.3 MB/s  0:00:00
Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
Downloading s3transfer-0.19.2-py3-none-any.whl (90 kB)
Using cached urllib3-2.8.0-py3-none-any.whl (135 kB)
Downloading deprecated-3.0.0-py3-none-any.whl (21 kB)
Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
Downloading gdown-6.4.2-py3-none-any.whl (50 kB)
Downloading jsonlines-4.0.0-py3-none-any.whl (8.7 kB)
Using cached attrs-26.1.0-py3-none-any.whl (67 kB)
Downloading lxml-6.1.3-cp312-cp312-manylinux_2_26_x86_64.manylinux_2_28_x86_6
4.whl (5.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.2/5.2 MB 108.9 MB/s  0:00:00
Downloading matplotlib-3.11.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17
_x86_64.whl (9.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.9/9.9 MB 103.3 MB/s  0:00:00
Downloading contourpy-1.4.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_
x86_64.whl (383 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.67.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_
x86_64.whl (5.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.4/5.4 MB 97.1 MB/s  0:00:00
Downloading kiwisolver-1.5.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_
x86_64.whl (1.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.5/1.5 MB 95.8 MB/s  0:00:00
Downloading more_itertools-11.2.1-py3-none-any.whl (74 kB)
Downloading mpld3-0.5.12-py3-none-any.whl (203 kB)
Downloading pillow-12.3.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x8
6_64.whl (6.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.9/6.9 MB 105.5 MB/s  0:00:00
Downloading pyparsing-3.3.3-py3-none-any.whl (126 kB)
Downloading pytorch_revgrad-0.2.0-py3-none-any.whl (4.6 kB)
Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86
_64.manylinux_2_28_x86_64.whl (807 kB)
Using cached regex-2026.9.29-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_
x86_64.manylinux_2_28_x86_64.whl (807 kB)
Using cached safetensors-0.8.0-cp310-abi3-manylinux_2_17_x86_64.manylinux2014
_x86_64.whl (516 kB)
Downloading scikit_learn-1.9.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_
28_x86_64.whl (9.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9.2/9.2 MB 99.6 MB/s  0:00:00
Using cached joblib-1.6.0-py3-none-any.whl (306 kB)
Using cached cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Downloading narwhals-2.27.1-py3-none-any.whl (483 kB)
Downloading segtok-1.5.11-py3-none-any.whl (24 kB)
Downloading sentencepiece-0.2.2-cp312-cp312-manylinux_2_27_x86_64.manylinux_2
_28_x86_64.whl (1.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.4/1.4 MB 107.7 MB/s  0:00:00
Downloading tabulate-0.10.0-py3-none-any.whl (39 kB)
Downloading threadpoolctl-3.7.0-py3-none-any.whl (26 kB)
Using cached tqdm-4.70.1-py3-none-any.whl (80 kB)
```

```
Downloading transformer_smaller_training_vocab-0.4.2-py3-none-any.whl (14 kB)
Downloading accelerate-1.15.0-py3-none-any.whl (394 kB)
Downloading wikipedia_api-0.16.0-py3-none-any.whl (129 kB)
Using cached click-8.5.0-py3-none-any.whl (125 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Downloading tenacity-9.1.4-py3-none-any.whl (28 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached anyio-4.15.1-py3-none-any.whl (132 kB)
Using cached idna-3.20-py3-none-any.whl (69 kB)
Using cached beautifulsoup4-4.15.0-py3-none-any.whl (109 kB)
Downloading soupsieve-3.0.3-py3-none-any.whl (41 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Downloading intervaltree-3.2.1-py2.py3-none-any.whl (25 kB)
Downloading protobuf-7.36.2-cp310-abi3-manylinux2014_x86_64.whl (343 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.5.2-cp312-cp312-manylinux2014_x86_64.manyli
nux_2_17_x86_64.manylinux_2_28_x86_64.whl (257 kB)
Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Building wheels for collected packages: langdetect, pptree, sqlitedict, docop
t
  Building wheel for langdetect (pyproject.toml) ... done
  Created wheel for langdetect: filename=langdetect-1.0.9-py3-none-any.whl si
ze=996861 sha256=283d19e38f19cd133834b371222461e92b323b59fcf486050c99228f713c
75f9
  Stored in directory: /home/damian/.cache/pip/wheels/c1/67/88/e844b5b022812e
15a52e4eaa38a1e709e99f06f6639d7e3ba7
  Building wheel for pptree (pyproject.toml) ... done
  Created wheel for pptree: filename=pptree-3.1-py3-none-any.whl size=4635 sh
a256=c45b57e06583764db9d5d9976ede2fc72e16875f6814902c306919b8573aa6fc
  Stored in directory: /home/damian/.cache/pip/wheels/b0/2d/de/37058114a8f07c
fec75747cb46b864bc5c71b0e9e0e4cd0acd
  Building wheel for sqlitedict (pyproject.toml) ... done
  Created wheel for sqlitedict: filename=sqlitedict-2.1.0-py3-none-any.whl si
ze=16871 sha256=8270c34cffc5f81ee96ca25db437a838c600d9736edb1933d5ab91b714569
4c1
  Stored in directory: /home/damian/.cache/pip/wheels/7a/6f/21/fc016aef45ffca
be27129a2252f061387cbf278d2086225a64
  Building wheel for docopt (pyproject.toml) ... done
  Created wheel for docopt: filename=docopt-0.6.2-py2.py3-none-any.whl size=1
3746 sha256=1a1288248ba26cb6ef280d7de3d5cf79496eeaf648ac7a618cdd6112e70f59e8
  Stored in directory: /home/damian/.cache/pip/wheels/1a/bf/a1/4cee4f7678c68c
5875ca89eaccf460593539805c3906722228
Successfully built langdetect pptree sqlitedict docopt
Installing collected packages: sqlitedict, sortedcontainers, pptree, docopt,
urllib3, tqdm, threadpoolctl, tenacity, tabulate, soupsieve, sentencepiece, s
afetensors, regex, pyyaml, PySocks, pyparsing, protobuf, pillow, narwhals, mo
re-itertools, lxml, langdetect, kiwisolver, jmespath, intervaltree, idna, hf-
xet, h11, ftfy, fonttools, deprecated, cycler, contourpy, conllu, cloudpickle
, click, charset_normalizer, certifi, attrs, segtok, requests, matplotlib, js
onlines, joblib, httpcore, botocore, beautifulsoup4, anyio, scikit-learn, s3t
ransfer, mpld3, huggingface-hub, httpx, bioc, wikipedia-api, tokenizers, gdow
n, boto3, transformers, pytorch-revgrad, accelerate, transformer-smaller-trai
ning-vocab, flair
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 63/63 [flair]m [flair]m [accelera
```

```
te]s]ub]
Successfully installed PySocks-1.7.1 accelerate-1.15.0 anyio-4.15.1 attrs-2
6.1.0 beautifulsoup4-4.15.0 bioc-2.1 boto3-1.43.114 botocore-1.43.114 certif
i-2026.7.22 charset_normalizer-3.5.2 click-8.5.0 cloudpickle-3.1.2 conll
u-4.5.3 contourpy-1.4.0 cycler-0.12.1 deprecated-3.0.0 docopt-0.6.2 flair-0.1
5.1 fonttools-4.67.0 ftfy-6.3.1 gdown-6.4.2 h11-0.16.0 hf-xet-1.7.0 httpcor
e-1.0.9 httpx-0.28.1 huggingface-hub-0.36.2 idna-3.20 intervaltree-3.2.1 jmes
path-1.1.0 joblib-1.6.0 jsonlines-4.0.0 kiwisolver-1.5.1 langdetect-1.0.9 lxm
l-6.1.3 matplotlib-3.11.2 more-itertools-11.2.1 mpld3-0.5.12 narwhals-2.27.1
pillow-12.3.0 pptree-3.1 protobuf-7.36.2 pyparsing-3.3.3 pytorch-revgra
d-0.2.0 pyyaml-6.0.3 regex-2026.9.29 requests-2.34.2 s3transfer-0.19.2 safete
nsors-0.8.0 scikit-learn-1.9.1 segtok-1.5.11 sentencepiece-0.2.2 sortedcontai
ners-2.4.0 soupsieve-3.0.3 sqlitedict-2.1.0 tabulate-0.10.0 tenacity-9.1.4 th
readpoolctl-3.7.0 tokenizers-0.22.2 tqdm-4.70.1 transformer-smaller-training-
vocab-0.4.2 transformers-4.57.6 urllib3-2.8.0 wikipedia-api-0.16.0

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
```

In [16]:
```
from flair.embeddings import FlairEmbeddings, StackedEmbeddings
from flair.data import Sentence
```

```
/home/damian/szkola/PJN_Laboratoria/lab_02/venv/lib64/python3.12/site-package
s/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter an
d ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.ht
ml
  from .autonotebook import tqdm as notebook_tqdm
```

In [17]:
```
embeddings = FlairEmbeddings('news-forward')

#tworzymy objekty Sentence
s1 = Sentence("How does it feel to be a scribe?")
s2 = Sentence("There is no such thing as simply good or bad")
sentences = [s1, s2]


# Obliczamy embeddingi(zanurzamy słowa)
embeddings.embed(sentences)

for sentence in sentences: #Zanurzone zdania nadal są obiektem Sentence
  print(sentence)
```

```
2026-10-15 12:47:25,517 https://flair.informatik.hu-berlin.de/resources/embed
dings/flair/news-forward-0.4.1.pt not found in cache, downloading to /tmp/tmp
3rl31jfl
100%|██████████| 69.7M/69.7M [00:02<00:00, 31.1MB/s]
2026-10-15 12:47:28,119 copying /tmp/tmp3rl31jfl to cache at /home/damian/.fl
air/embeddings/news-forward-0.4.1.pt
2026-10-15 12:47:28,142 removing temp file /tmp/tmp3rl31jfl
Sentence[9]: "How does it feel to be a scribe?"
Sentence[10]: "There is no such thing as simply good or bad"
```

In [18]:
```
#dostajemy embeddingi poszczególnych słów
for sentence in sentences:
  for token in sentence:
    print(token.embedding)
```

```
#embeddingi są tensorami
```

```
tensor([-0.0011, -0.0021,  0.0038,  ..., -0.0004,  0.0099,  0.0547],
       device='cuda:0')
tensor([-0.0016, -0.0261,  0.0389,  ...,  0.0031, -0.0070,  0.0025],
       device='cuda:0')
tensor([0.0174, 0.0181, 0.0148,  ..., 0.0051, 0.0051, 0.0130], device='cud
a:0')
tensor([ 0.0001,  0.0128,  0.0476,  ..., -0.0084, -0.0023,  0.0106],
       device='cuda:0')
tensor([-2.1703e-03,  8.3835e-05,  2.5111e-02,  ..., -2.8422e-04,
        -1.2085e-02,  2.9504e-02], device='cuda:0')
tensor([-0.0032,  0.0020,  0.0281,  ...,  0.0004, -0.0017, -0.0106],
       device='cuda:0')
tensor([ 5.3354e-03,  3.8057e-05, -2.2038e-02,  ..., -4.1645e-04,
         1.9285e-02, -1.2343e-02], device='cuda:0')
tensor([-1.0603e-03,  4.8692e-05,  2.1461e-02,  ..., -1.1519e-05,
        -4.3333e-03,  3.1237e-03], device='cuda:0')
tensor([ 8.7278e-04,  3.7830e-06, -4.9728e-04,  ..., -1.4234e-04,
         5.4650e-03,  3.0912e-03], device='cuda:0')
tensor([-0.0012, -0.0002, -0.1499,  ...,  0.0024,  0.0017,  0.1649],
       device='cuda:0')
tensor([ 0.0078, -0.0047,  0.0564,  ..., -0.0012,  0.0325,  0.0035],
       device='cuda:0')
tensor([ 5.8101e-02, -7.9280e-05,  9.8927e-02,  ..., -2.7515e-03,
         8.8246e-04,  2.6967e-02], device='cuda:0')
tensor([-0.0004, -0.0036,  0.0782,  ..., -0.0004, -0.0344, -0.0005],
       device='cuda:0')
tensor([ 0.0015,  0.0033, -0.0119,  ..., -0.0005, -0.0164,  0.0293],
       device='cuda:0')
tensor([ 1.2548e-02, -1.0575e-03, -1.1854e-02,  ..., -2.9870e-05,
         4.1029e-03,  2.8729e-02], device='cuda:0')
tensor([-0.0053, -0.0011,  0.0058,  ..., -0.0018,  0.0212,  0.0082],
       device='cuda:0')
tensor([ 2.0066e-04,  9.6434e-04, -2.2161e-02,  ...,  6.8256e-05,
        -1.3960e-01,  3.5031e-03], device='cuda:0')
tensor([-9.1947e-05, -1.7716e-03,  1.0522e-02,  ..., -1.9223e-03,
        -1.2772e-05,  3.2008e-02], device='cuda:0')
tensor([-0.0006,  0.0004, -0.0019,  ..., -0.0005,  0.0061, -0.0073],
       device='cuda:0')
```

In [19]:
```
import numpy as np
```

In [21]:
```
for sentence in sentences:
  for token in sentence:
    print(np.array(token.embedding.cpu().numpy()))
```

```
[-0.00108336 -0.00213309  0.00378132 ... -0.00035749  0.00994315
  0.05468404]
[-0.00163753 -0.02605824  0.03892935 ...  0.00308809 -0.00699125
  0.00248131]
[0.01742223 0.01808094 0.01483307 ... 0.00506839 0.0050871  0.01303806]
[ 0.00013388  0.01283137  0.04761452 ... -0.0083766  -0.00231394
  0.0105772 ]
[-2.1703204e-03  8.3835272e-05  2.5111243e-02 ... -2.8422478e-04
 -1.2084963e-02  2.9504092e-02]
[-0.00319963  0.00195649  0.02806855 ...  0.00040728 -0.00174367
 -0.01060929]
[ 5.3354464e-03  3.8057275e-05 -2.2037642e-02 ... -4.1644566e-04
  1.9284626e-02 -1.2343276e-02]
[-1.0603219e-03  4.8691731e-05  2.1461276e-02 ... -1.1519401e-05
 -4.3332502e-03  3.1237055e-03]
[ 8.7277684e-04  3.7830009e-06 -4.9727742e-04 ... -1.4234209e-04
  5.4650400e-03  3.0911700e-03]
[-0.00118221 -0.00018428 -0.14988717 ...  0.00243919  0.00166817
  0.16486977]
[ 0.00784009 -0.00474323  0.05644518 ... -0.00120977  0.03249407
  0.00351099]
[ 5.8100540e-02 -7.9280333e-05  9.8926567e-02 ... -2.7514719e-03
  8.8246464e-04  2.6966682e-02]
[-0.00039669 -0.00361356  0.07815546 ... -0.00039067 -0.03442113
 -0.00054164]
[ 0.00145081  0.00325835 -0.011852   ... -0.00050557 -0.01640288
  0.02930536]
[ 1.2548085e-02 -1.0575150e-03 -1.1854418e-02 ... -2.9870203e-05
  4.1028536e-03  2.8728830e-02]
[-0.0052818  -0.00108876  0.00579742 ... -0.00176392  0.02116122
  0.00819341]
[ 2.0066126e-04  9.6433813e-04 -2.2161406e-02 ...  6.8255991e-05
 -1.3960406e-01  3.5030779e-03]
[-9.1946589e-05 -1.7716442e-03  1.0521732e-02 ... -1.9222908e-03
 -1.2771503e-05  3.2008205e-02]
[-0.00058447  0.00037403 -0.00193827 ... -0.00045992  0.00614226
 -0.00726807]
```

Embeddingi dla zdań

In [22]:
```
from flair.embeddings import DocumentPoolEmbeddings
```

In [23]:
```
sentence_embeddings = DocumentPoolEmbeddings([FlairEmbeddings('news-forward'
```

In [24]:
```
sentence_embeddings.embed(sentences)
```

In [25]:
```
for sentence in sentences:
  print(sentence.embedding)
```

```
tensor([ 0.0016,  0.0005,  0.0175,  ..., -0.0001,  0.0014,  0.0104],
       device='cuda:0')
tensor([ 0.0073, -0.0008,  0.0052,  ..., -0.0007, -0.0124,  0.0289],
       device='cuda:0')
```

Zadania

Zadanie1

Załaduj polski model kontekstowych embeddingów Flair:

In [27]:
```
embeddings = FlairEmbeddings('pl-forward')
```

```
2026-10-15 13:05:00,126 https://flair.informatik.hu-berlin.de/resources/embeddings/flair/lm-polish-forward-v0.2.pt not found in cache, downloading to /tmp/tmpgdazkex9
100%|██████████| 80.3M/80.3M [00:02<00:00, 33.1MB/s]
2026-10-15 13:05:02,917 copying /tmp/tmpgdazkex9 to cache at /home/damian/.flair/embeddings/lm-polish-forward-v0.2.pt
2026-10-15 13:05:02,951 removing temp file /tmp/tmpgdazkex9
```

In [28]:
```
s1 = Sentence("Zamek w Malborku jest bardzo stary.")
s2 = Sentence("Suwak w kurtce to zamek błyskawiczny")
s3 = Sentence("Sprawdź, czy zamek w drzwiach działa")
sentences = [s1, s2, s3]
```

Oblicz embeddingi dla wszystkich zdań:

In [29]:
```
embeddings.embed(sentences)
```

Out[29]:
```
[Sentence[7]: "Zamek w Malborku jest bardzo stary.",
 Sentence[6]: "Suwak w kurtce to zamek błyskawiczny",
 Sentence[7]: "Sprawdź, czy zamek w drzwiach działa"]
```

Wypisz każde zdanie i dla słowa "zamek", wypisz pierwsze 5 wymiarów embeddingu:

In [35]:
```
for sentence in sentences:
    #TODO
    for token in sentence:
        if token.text.lower() == "zamek":
            print(f'{sentence} | embedding: {token.embedding.cpu().numpy()[:
```

```
Sentence[7]: "Zamek w Malborku jest bardzo stary." | embedding: [-5.4497854
e-03 -2.1578195e-02  4.5388409e-05 -1.7060013e-04
 -4.3132866e-04]
Sentence[6]: "Suwak w kurtce to zamek błyskawiczny" | embedding: [-0.00433285
-0.01309874  0.00041927 -0.00013573 -0.00035192]
Sentence[7]: "Sprawdź, czy zamek w drzwiach działa" | embedding: [-4.2989510
e-03 -2.1091273e-02  7.7383994e-04 -6.9892907e-05
 -3.6811165e-04]
```

Czy embeddingi są różne w zależności od znaczenia słowa?

Odpowiedź: Tak

Zadanie 2

Stwórz bota z bazą przykładownych pytań i odpowiedzi, zwracającego odpowiedź na zapytanie na podstawie podobieństwa (WMD albo flair document embeddings) pomiędzy zapytniem użytownika a pytaniami w swojej bazie

Przykład:

Zbiór pytań i odpowiedzi :

```
[
  {"q1" : "a1"},
  {"q2" : "a2"},
  {"q3" : "a3"},
  ...
  {"qn" : "an"},
]
```

Wektoryzujemy pytania:

```
  {"q1_vecotorized" : "a1"},
  {"q2_vecotorized" : "a2"},
  {"q3_vecotorized" : "a3"},
  ...
  {"qn_vecotorized" : "an"},
```

Dla pytania użytkownia pytania q_user wektoryzujemy je i liczmy podobieństwo z pytaniami w bazie

```
sim("q_user_vectorized","q1_vecotorized"),
sim("q_user_vectorized","q2_vecotorized"), ...
("q_user_vectorized""qn_vecotorized")
```

Zwracamy odpowiedź do pytania z bazy najbardziej podobnego do pytania użytkownika

$$argmax(sim(q_{user}vectorized, q_ivecotorized))$$

Do liczenia podobnieństma można użyć np https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cosine.html ale wtedy robimy argmin bo

$$cosine\ distance = 1 - cosine\ similarity$$

In [90]:
```
qa = [
  {"q": "Jak mogę zapisać się na zajęcia?", "a": "Zapisy odbywają się przez
  {"q": "Gdzie znajdę plan zajęć?", "a": "Plan zajęć jest dostępny w serwisi
  {"q": "Jak zaliczyć przedmiot?", "a": "Zaliczenie zależy od prowadzącego,
  {"q": "Co zrobić, gdy nie zdam egzaminu?", "a": "Możesz przystąpić do egza
```

```
  {"q": "Kiedy zaczyna się sesja egzaminacyjna?", "a": "Sesja zaczyna się po
  {"q": "Jak uzyskać stypendium naukowe?", "a": "Trzeba złożyć wniosek w dzi
  {"q": "Gdzie mogę wypożyczyć książki?", "a": "Książki wypożyczysz w biblio
  {"q": "Jak skontaktować się z prowadzącym zajęcia?", "a": "Napisz na uczel
]

qa_vectorized = []
for sentence in qa:
    s = Sentence(sentence["q"])
    sentence_embeddings.embed(s)
    qa_vectorized.append({"q": s.embedding, "a": sentence["a"]})

print(qa_vectorized)
```

```
[{'q': tensor([ 9.5746e-04, -8.7132e-05,  1.3748e-03,  ..., -6.4646e-04,
         8.6292e-03,  2.2359e-02], device='cuda:0'), 'a': 'Zapisy odbywają si
ę przez system dziekanatu w terminie ogłoszonym na początku semestru.'},
{'q': tensor([ 0.0014,  0.0002, -0.0074,  ...,  0.0003, -0.0003,  0.0062],
       device='cuda:0'), 'a': 'Plan zajęć jest dostępny w serwisie uczelni w
zakładce Plan zajęć.'}, {'q': tensor([ 7.2596e-04,  1.9473e-06,  5.4327e-03,
..., -3.4837e-04,
         1.2773e-02,  8.9810e-03], device='cuda:0'), 'a': 'Zaliczenie zależy
od prowadzącego, zwykle jest to kolokwium, projekt albo egzamin końcowy.'},
{'q': tensor([ 0.0010,  0.0001,  0.0267,  ..., -0.0005,  0.0135,  0.0281],
       device='cuda:0'), 'a': 'Możesz przystąpić do egzaminu poprawkowego w t
erminie ustalonym przez prowadzącego.'}, {'q': tensor([ 1.7145e-04, -3.6697
e-05,  2.1302e-03,  ...,  4.9880e-04,
         1.0953e-02,  2.5746e-02], device='cuda:0'), 'a': 'Sesja zaczyna się
po zakończeniu zajęć w semestrze, dokładne daty są w harmonogramie roku akade
mickiego.'}, {'q': tensor([ 7.3467e-04, -1.8011e-05,  1.4132e-02,  ..., -7.98
29e-04,
         8.9412e-03,  1.9061e-02], device='cuda:0'), 'a': 'Trzeba złożyć wnio
sek w dziekanacie i spełnić wymaganie dotyczące średniej ocen.'}, {'q': tenso
r([ 0.0007,  0.0003, -0.0005,  ..., -0.0008, -0.0082,  0.0094],
       device='cuda:0'), 'a': 'Książki wypożyczysz w bibliotece uczelni po ok
azaniu legitymacji studenckiej.'}, {'q': tensor([ 0.0007, -0.0001, -0.0112,
..., -0.0004,  0.0244,  0.0271],
       device='cuda:0'), 'a': 'Napisz na uczelniany adres e-mail prowadzącego
albo przyjdź na jego konsultacje.'}]
```

In [99]:

```
# importy
from scipy.spatial.distance import cosine

def bot(user_question):
  # TODO
  user_q_sentence = Sentence(user_question)
  sentence_embeddings.embed(user_q_sentence)
  sim = [calculate_similarity(user_q_sentence.embedding.cpu().numpy(), s["q"
  sim_id = np.argmax(sim)

  print(f"Pytanie użytkownika: {user_question}")
  print(f"Najbardziej podobne pytanie z bazy danych: {qa[sim_id]["q"]}")
  print(f"Odpowiedź: {qa_vectorized[sim_id]["a"]}")

def calculate_similarity(vec1, vec2):
    return 1 - cosine(vec1, vec2)
```

In [101…
```
# TODO - zadaj 3 pytania
bot("Jaki jest kontakt do prowadzącego?")
print()
bot("Co trzeba zrobić, aby uzyskać stypendium?")
print()
bot("Gdzie jest plan zajęć?")
```

```
Pytanie użytkownika: Jaki jest kontakt do prowadzącego?
Najbardziej podobne pytanie z bazy danych: Jak skontaktować się z prowadzącym zajęcia?
Odpowiedź: Napisz na uczelniany adres e-mail prowadzącego albo przyjdź na jego konsultacje.

Pytanie użytkownika: Co trzeba zrobić, aby uzyskać stypendium?
Najbardziej podobne pytanie z bazy danych: Co zrobić, gdy nie zdam egzaminu?
Odpowiedź: Możesz przystąpić do egzaminu poprawkowego w terminie ustalonym przez prowadzącego.

Pytanie użytkownika: Gdzie jest plan zajęć?
Najbardziej podobne pytanie z bazy danych: Gdzie znajdę plan zajęć?
Odpowiedź: Plan zajęć jest dostępny w serwisie uczelni w zakładce Plan zajęć.
```